

Yi-Chao Chen (陳奕超)

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Room 505, Shanghai Jiao Tong University,
Shanghai, 200240

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EXPERIENCE

- Associate Professor**, *Shanghai Jiao Tong University, China* Sept. 2018 ~ Now
Research Topic: Mobile Computing, Multimodal Sensing, Cybersecurity
- Visiting Scholar**, *National Institute of Informatics, Tokyo, Japan* Aug. 2024 ~ Oct. 2024
Research Topic: Acoustic Sensing and Synthesis
- Visiting Scholar**, *Microsoft Research Asia, Shanghai, China* Oct. 2023 ~ June 2024
Research Topic: Multimodal Sensing and Alignment
- Chief Scientist**, *Hauoli LLC, TX, USA* Feb. 2017 ~ Sept. 2018
Research Topic: Human-machine Collaboration
- Researcher**, *Huawei, HongKong* Sept. 2015 ~ Feb. 2017
Research Topic: Software Defined Networks, Internet of Things
- Research Intern**, *AT&T Research Labs, NJ, USA* June 2011 ~ Aug. 2011
Mentor: Dr. Jia Wang, Dr. Nicholas Duffield.
Research Topic: Anomaly Detection, Social Networks
- Research Assistant**, *Academia Sinica, Taipei, Taiwan* Jan. 2008 ~ July 2009
Mentor: Dr. Ling-Jyh Chen.
Research Topic: Delay-Tolerant Networking (DTN)

EDUCATION

- The University of Texas at Austin**, *Texas, USA* Aug. 2009 ~ Sept 2015
Ph.D. in Computer Science
- National Taiwan University**, *Taipei, Taiwan* Sept. 2004 ~ June 2006
M.S. in Computer Science and Information Engineering
- National Taiwan University**, *Taipei, Taiwan* Sept. 2000 ~ June 2004
B.S. in Computer Science and Information Engineering

SELECTED PUBLICATION

Conference Papers

- Lanqing Yang, Weiyi Wang, Yongzhao Zhang, Yixin Liu, Yi-Chao Chen^{*}, Guangtao Xue, Ahmad Ali. *SingSprite: Non-Tactile Appliance Interaction for the Blind Through Power-Supply Acoustic Signatures*. ACM **SenSys** 2026.
- Runting Zhang, Yijie Li, Dian Ding, Yi-Chao Chen^{*}, Yida Wang, Dongyao Chen, Jingxian Wang, Jiadi Yu, Ling Ma, and Guangtao Xue. *Bridge: Enabling BLE Direction Finding Feature Compatible with All Bluetooth Devices*. ACM **MobiCom** 2025.
- Hao Pan, Yezhou Wang, Jiting Liu, Ruichun Ma, Lili Qiu, Yi-Chao Chen^{*}, Guangtao Xue, and Ju Ren. *CGMM: Non-Invasive Continuous Glucose Monitoring in Wearables Using Metasurfaces*. ACM **MobiCom** 2025.
- Yezhou Wang, Hao Pan^{*}, Yongjian Fu, Qinyun Hu, Lili Qiu, Yi-Chao Chen, Guangtao Xue, Ju Ren. *WDNN*:

Weighted Diffractive Neural Network for Physical-layer RF Signal Processing. ACM **MobiCom** 2025.

- Yida Wang, Yu Lu, Yuxuan ZHOU, Yifei Shen, Lili Qiu, Zeyuan Lai, Yi-Chao Chen*, Hao Pan, Juntao Zhou, Dian Ding, Mei Wang, Guangtao Xue, and Qian Zhang. *High-resolution mmWave Imaging using Metasurface and Diffusion*. ACM **MobiSys** 2025.
- Juntao Zhou, Dian Ding*, Yijie Li, Yu Lu, Yida Wang, Yongzhao Zhang, Yi-Chao Chen*, and Guangtao Xue. *M²SILENT: Enabling Multi-user Silent Speech Interactions via Multi-directional Speakers in Shared Spaces*. **CHI** 2025.
- Runtong Zhang, Yijie Li*, Dian Ding, Hao Pan, Yongzhao Zhang, Yi-Chao Chen*, Xiaoyu Ji, Jiadi Yu, Guangtao Xue. *SADIF: Spoofing Attack on BLE Direction Finding Based Localization System*. ACM **MobiHoc** 2025.
- Yu Lu, Dian Ding, Hao Pan*, Jiatong Ding, Yongjian Fu, Yi-Chao Chen*, Ju Ren, Guangtao Xue. *MODepth: Benchmarking Multi-frame Monocular Depth Estimation with Optical Image Stabilization*. **SIGGRAPH Asia** 2025.
- Yu Lu, Ran Wang, Dian Ding, Han Zhang, Liyun Zhang, Lanqing Yang, Yi-Chao Chen*, and Guangtao Xue. *AMSER: Accelerate Mobile Speech Emotion Recognition with Signal Compression*. IEEE **ICASSP** 2025.
- Haonan Wu, Erci Xu*, Ligang Wang, Yuandong Hong, Changsheng Niu, Bo Shi, Lingjun Zhu, Jinnian He, Dong Wu, Weidong Zhang, Qiuping Wang, Changhong Wang, Xinqi Chen, Guangtao Xue, Yi-Chao Chen, Dian Ding. *Hey Hey, My My, Skewness Is Here to Stay: Challenges and Opportunities in Cloud Block Store Traffic*. **EuroSys** 2025.
- Yu Lu, Dian Ding, Hao Pan, Yongjian Fu, Liyun Zhang, Feitong Tan, Ran Wang, Yi-Chao Chen*, Guangtao Xue, Ju Ren. *M3Cam: Extreme Super-resolution via Multi-Modal Optical Flow for Mobile Cameras*. ACM **SenSys** 2024.
- Yongjian Fu, Yongzhao Zhang, Yu Lu, Lili Qiu, Yi-Chao Chen*, Yezhou Wang, Mei Wang, Yijie Li, Ju Ren, Yaoyue Zhang. *Adaptive Metasurface-Based Acoustic Imaging using Joint Optimization*. ACM **MobiSys** 2024.
- Yunzhong Chen, Jiadi Yu*, Yingying Chen, Linghe Kong, Yanmin Zhu, Yi-Chao Chen. *RFSpY: Eavesdropping on Online Conversations with Out-of-Vocabulary Words by Sensing Metal Coil Vibration of Headsets Leveraging RFID*. ACM **MobiSys** 2024.
- Yezhou Wang, Hao Pan, Lili Qiu, Linghui Zhong, Jiting Liu, Ruichun Ma, Yi-Chao Chen*, Guangtao Xue, and Ju Ren. *GPMS: Enabling Indoor GNSS Positioning using Passive Metasurfaces*. ACM **MobiCom** 2024.
- Yiwen Song, Hao Pan*, Longyuan Ge, Lili Qiu, Swarun Kumar*, Yi-Chao Chen. *Guiding Energy Distribution inside Microwave Oven*. ACM **MobiCom** 2024.
- Yijie Li, Juntao Zhou, Dian Ding*, Yi-Chao Chen*, Guangtao Xue, Jiadi Yu, Lili Qiu. *MuDiS: An Audio-independent, Wide-angle, and Leak-free Multi-directional Speaker*. ACM **MobiCom** 2024.
- Yu Lu, Juntao Zhou, Yongjian Fu, Yijie Li, Dian Ding, Hao Pan, Yongzhao Zhang, Yi-Chao Chen*, and Guangtao Xue. *HandPad: Make Your Hand an On-the-go Writing Pad via Human Capacitance*. **UIST** 2024.
- Yuehu Jiang, Haonan Wu, Yi-Chao Chen*, and Guangtao Xue. *High Speed and Low Cost One-to-Many VLC Using Polymer-Dispersed Liquid Crystals*. IEEE **ICC** 2024.
- Hao Pan, Lili Qiu, Bei Ouyang, Yongzhao Zhang, Shicheng Zheng, Yi-Chao Chen*, Guangtao Xue. *AMSat: Adaptive Metasurface for Low Earth Orbit Satellite Communication*. ACM **MobiCom** 2023.
- Yongzhao Zhang, Yezhou Wang, Lanqing Yang, Mei Wang, Yi-Chao Chen*, Lili Qiu*, Yihong Liu, Guangtao Xue, and Jiadi Yu. *Acoustic Sensing and Communication Using Metasurface*. USENIX **NSDI** 2023.
- Yongzhao Zhang, Hao Pan, Yi-Chao Chen*, Lili Qiu, Yu Lu, Guangtao Xue, Jiadi Yu, Feng Lyu, and Haonan Wang. *Addressing Practical Challenges in Acoustic Sensing To Enable Fast Motion Tracking*. ACM **IPSN** 2023.
- Yu Lu, Hao Pan, Feitong Tan, Yi-Chao Chen*, Jiadi Yu, Jinghai He, and Guangtao Xue. *Effectively Learning Moiré QR Code Decryption from Simulated Data*. IEEE **Infocom** 2023.
- Lanqing Yang, Xinqi Chen, Xiangyong Jian, Leping Yang, Yijie Li, Qianfei Ren, Yi-Chao Chen*, Guangtao Xue, and Xiaoyu Ji. *Remote Attacks on Speech Recognition Systems Using Sound from Power Supply*. USENIX

Security 2023.

- Hao Pan, Feitong Tan, Wenhao Li, Yi-Chao Chen^{*}, and Guangtao Xue. *OISSR: Optical Image Stabilization Based Super Resolution on Smartphone Cameras*. ACM **MM** 2022.
- Hao Pan, Feitong Tan, Yi-Chao Chen^{*}, Gaoang Huang, Qingyang Li, Wenhao Li, Guangtao Xue, Lili Qiu, and Xiaoyu Ji. *DoCam: depth sensing with an optical image stabilization supported RGB camera*. ACM **MobiCom** 2022.
- Hao Kong, Xiangyu Xu, Jiadi Yu^{*}, Qilin Chen, Chenguang Ma, Yingying Chen, Yi-Chao Chen, and Linghe Kong. *M3Track: mmWave-Based Multi-User 3D Posture Tracking*. ACM **MobiSys** 2022.
- Hao Pan, Feitong Tan, Wenhao Li, Yi-Chao Chen^{*}, Lanqing Yang, Guangtao Xue, and Xiaoyu Ji. *MagDefender: Detecting Eavesdropping on Mobile Devices using the Built-in Magnetometer*. IEEE **SECON** 2022.
- Min-Wei Hung, Chien Wen (Tina) Yuan, Nanyi Bi, Yi-Chao Chen^{*}, Wan-Chen Lee, Ming-Chyi Huang, Chuang-Wen You^{*}. *To Use or Abuse: Opportunities and Difficulties in the Use of Multi-channel Support to Reduce Technology Abuse by Adolescents*. ACM **CSCW** 2022.
- Haonan Wu, Yi-Chao Chen^{*}, Yuehu Jiang, Guangtao Xue, Ming Wang, Shiyu Qian, Jiadi Yu, Pai-Yen Chen. *OnionCode: Enabling Multi-priority Coding in LED-based Optical Camera Communications*. IEEE **Infocom** 2022.
- Hao Kong, Li Lu, Jiadi Yu^{*}, Yingying Chen, Xiangyu Xu, Feilong Tang, Yi-Chao Chen. *MultiAuth: Enable Multi-User Authentication with Single Commodity WiFi Device*. ACM **MobiHoc** 2021.
- Hao Pan, Yi-Chao Chen^{*}, Qi Ye, Guangtao Xue. *MagicInput: Training-free Multi-lingual Finger Input System using Data Augmentation based on MNISTs*. **IPSN** 2021.
- Yushi Cheng, Xiaoyu Ji^{*}, Lixu Wang, Qi Pang, Yi-Chao Chen, Wenyan Xu. *mID: Tracing Screen Photos via Moiré Patterns*. USENIX **Security** 2021.
- Yijie Li, Yi-Chao Chen^{*}, Xiaoyu Ji, Hao Pan, Lanqing Yang, Guangtao Xue, Jiadi Yu. *ScreenID: Enhancing QRCode Security by Fingerprinting Screens*. IEEE **Infocom** 2021.
- Lanqing Yang, Yi-Chao Chen^{*}, Hao Pan, Dian Ding, Guangtao Xue, Linghe Kong, Jiadi Yu, Minglu Li. *MagPrint: Deep Learning Based User Fingerprinting Using Electromagnetic Signals*. IEEE **Infocom** 2020.
- Xiangyu Xu, Jiadi Yu^{*}, Yingying Chen, Qin Hua, Yanmin Zhu, Yi-Chao Chen, Minglu Li. *TouchPass: Towards Behavior-irrelevant on-touch User Authentication on Smartphones Leveraging Vibrations*. ACM **MobiCom** 2020.
- Hao Pan, Lanqing Yang, Yi-Chao Chen^{*}, Guangtao Xue, Chuang-Wen You, Xiaoyu Ji. *mQRCode: Secure QR Code Using Nonlinearity of Spatial Frequency in Light*. ACM **MobiCom** 2019.
- Wenguang Mao, Mei Wang, Wei Sun, Lili Qiu, Swadhin Pradhan, Yi-Chao Chen. *RNN-based Room Scale Hand Motion Tracking*. ACM **MobiCom** 2019.
- Yushi Cheng, Xiaoyu Ji^{*}, Wenyan Xu, Yi-Chao Chen. *DeMiCPU: Device Fingerprinting with Magnetic Signals Radiated by CPU*. ACM **CCS** 2019.
- Hao Pan, Yi-Chao Chen^{*}, Guangtao Xue, Xiaoyu Ji. *MagneComm: Magnetometer-based Near-Field Communication*. ACM **MobiCom** 2017.
- Qun Huang, Xin Jin, Patrick P. C. Lee^{*}, Runhui Li, Lu Tang, Yi-Chao Chen, Gong Zhang. *SketchVisor: Robust Network Measurement for Software Packet Processing*. ACM **SigComm** 2017.
- Sangki Yun[#], Yi-Chao Chen[#], Huihuang Zheng, Lili Qiu, and Wenguang Mao. *Strata: Fine-Grained Device-Free Tracking Using Acoustic Signals*. ACM **MobiSys** 2017.
- Apurv Bhartiya[#], Yi-Chao Chen[#], George P. Nychis, Lili Qiu. *Embracing Distributed MIMO in Wireless Mesh Networks*. IEEE **ICNP** 2015.
- Sangki Yun[#], Yi-Chao Chen[#], Lili Qiu. *Turning a Mobile Device into a Mouse in the Air*. ACM **MobiSys** 2015.

- Yi-Chao Chen, Lili Qiu, Yin Zhang, Zhenxian Hu, Guangtao Xue. *Robust Network Compressive Sensing*. ACM **MobiCom** 2014.
- Yi-Chao Chen, Yong Liao, Mario Baldi, Sung-Ju Lee, Lili Qiu. *OS Fingerprinting and Tethering Detection in Mobile Networks*. ACM **IMC** 2014.
- Apurv Bhartia[#], Yi-Chao Chen[#], Swati Rallapalli, and Lili Qiu. *Harnessing Frequency Diversity in Multicarrier Wireless Networks*. ACM **MobiCom** 2011.
- Upendra Shevade, Yi-Chao Chen, Lili Qiu, Yin Zhang, Vinoth Chandar, Mi Kyung Han, Han Hee Song, Yousuk Seung. *Enabling high-bandwidth vehicular content distribution*. ACM **CoNEXT** 2010.
- Swati Rallapalli, Lili Qiu, Yin Zhang, and Yi-Chao Chen. *Exploiting Temporal Stability and Low Rank Structure for Mobile Network Localization*. ACM **MobiCom** 2010.

Journal Articles

- Yongzhao Zhang, Hao Pan, Dian Ding, Yue Pan, Yi-Chao Chen^{*}, Lili Qiu, Guangtao Xue, Ting Chen, and Xiaosong Zhang. *SwiftTrack+: Fine-Grained and Robust Fast Hand Motion Tracking Using Acoustic Signal*. IEEE/ACM **ToN** 2025.
- Yiwen Song, Hao Pan^{*}, Longyuan Ge, Lili Qiu, Swarun Kumar, and Yi-Chao Chen. *MicroSurf: Guiding Energy Distribution inside Microwave Oven with Metasurfaces*. ACM **GetMobile** 2025.
- Yongjian Fu, Lanqing Yang, Hao Pan, Yi-Chao Chen, Guangtao Xue, and Ju Ren^{*}. *MagSpy: Revealing User Privacy Leakage via Magnetometer on Mobile Devices*. IEEE **TMC** 2025.
- Hao Pan, Yongjian Fu, Ye Qi, Yi-Chao Chen^{*}, and Ju Ren^{*}. *MagicWrite: One-Dimensional Acoustic Tracking-Based Air Writing System*. IEEE **TMC** 2025.
- Guangtao Xue, Yijie Li, Hao Pan, Lanqing Yang, Yi-Chao Chen^{*}, Xiaoyu Ji, and Jiadi Yu. *ScreenID: Enhancing QRCode Security by Utilizing Screen Dimming Feature*. IEEE/ACM **ToN** 2022.
- Juntao Zhou, Dian Ding^{*}, Yu Lu, Yida Wang, Yijie Li, Yi-Chao Chen^{*}, and Guangtao Xue. *VISAR: Projecting Virtual Sound Spots for Acoustic Augmented Reality Using Air Nonlinearity*. **IMWUT** 2024.
- Guangtao Xue, Yijie Li, Hao Pan, Lanqing Yang, Yi-Chao Chen^{*}, Xiaoyu Ji, and Jiadi Yu. *ScreenID: Enhancing QRCode Security by Utilizing Screen Dimming Feature*. IEEE/ACM **ToN** 2022.
- Dian Ding, Lanqing Yang, Yi-Chao Chen^{*}, Guangtao Xue. *Handwriting Recognition System Leveraging Vibration Signal on Smartphones*. IEEE **TMC** 2022.
- Dian Ding, Lanqing Yang, Yi-Chao Chen^{*}, Guangtao Xue^{*}. *Leakage or Identification: Behavior-irrelevant User Identification Leveraging Leakage Current on Laptops*. ACM **IMWUT** 2021.
- Guangtao Xue, Hao Pan, Yi-Chao Chen^{*}, Xiaoyu Ji, Jiadi Yu. *MagneComm+: Near-Field Electromagnetic Induction Communication with Magnetometer*. IEEE **TMC** 2021.
- Xiaoyu Ji, Yushi Cheng, Wenyan Xu, Yuehan Chi, Hao Pan, Zhuangdi Zhu, Chuang-Wen You, Yi-Chao Chen, Lili Qiu. *No Seeing is Also Believing: Electromagnetic-emission-based Application Guessing Attacks via Smartphones*. IEEE **TMC** 2021.
- Yongzhao Zhang, Wei-Hsiang Huang, Chih-Yun Yang, Wen-Ping Wang, Yi-Chao Chen^{*}, Chuang-Wen You, Da-Yuan Huang, Guangtao Xue^{*}, Jiadi Yu. *Endophasia: Utilizing Acoustic-Based Imaging for Dropping Contact-Free Silent Speech Commands*. ACM **IMWUT** 2020.
- Liang Zhu, Mohamed Farhat, Yi-Chao Chen, Khaled N Salama, Pai-Yen Chen^{*}. *A Compact, Passive Frequency-Hopping Harmonic Sensor Based on a Microfluidic Reconfigurable Dual-Band Antenna*. IEEE **Sensors** 2020.

ACADEMIC SERVICES

TPC: ACM MobiCom 2025/2026, ACM MobiSys 2025/2026, ACM CoNEXT 2023/2024/2025/2026, EAI MobiQuitous 2026, IEEE ICPADS 2025.

Organization Committee: MIMSVAI (UbiComp Workshop) Co-chair 2021~2025, CoNEXT Publicity Chair 2024,

VNC Student Competition Chair 2018, 2020.

Secretary-General, Technical Committee on Computer Networks, Shanghai Computer Federation (2024~2027)

TEACHING

Foundations of Generative AI , short course, <i>Shanghai Jiao Tong University</i>	Spring 2025 ~ Now
Mobile Sensing and Computing , <i>Shanghai Jiao Tong University</i> (SJTU Student-rated A1~A2 class, top 5%~15%)	Fall 2018 ~ Now
Computer Networks , <i>Shanghai Jiao Tong University</i> (SJTU Student-rated A1~A2 class, top 5%~15%)	Fall 2018 ~ Now
The Technologies and Trend of IoT , <i>Shanghai Jiao Tong University</i> (SJTU MEM Program, Outstanding Teaching Award)	Spring 2019 ~ Now
CS 429: Computer Organization and Architecture , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Robert F. Dickerson	Spring 2015
CS 329E: Elements of Mobile Computing , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Robert F. Dickerson	Fall 2014
CS 361C: Information Assurance and Security , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Bill Young	Spring 2014
CS 337: Theory in Programming Practice , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Jayadev Misra	Spring 2013
CS 337: Theory in Programming Practice , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Mary Eberlein	Spring 2012
CS 395T: Learning Theory , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Adam Klivans and Dr. Pradeep Ravikumar	Fall 2011
CS 307: Foundations of Computer Science , <i>University of Texas at Austin</i> Teaching Assistant. Lecturer: Dr. Mike Scott	Fall 2010
Database Systems / Practicum in Database Systems , <i>National Taiwan University</i> Teaching Assistant. Lecturer: Dr. Hao-hua Chu	Fall 2005

HONOR AND AWARD

2025	ACM MobiSys 2025 Best Artifact Runner-Up.
2025	23rd China MBA Entrepreneurship Competition, Silver Award (Ranked 2nd out of over 500 teams). (第23屆中國MBA創業大賽、銀獎--超過500組參賽隊伍中排名第2).
2025	9th Shanghai Engineering Management Innovation Competition, Outstanding Faculty Advisor. (第九屆上海市工程管理創新大賽、優秀指導教師)
2024	IEEE ComSoc Student Competition, Honorary Mention (Advisor: Yi-Chao Chen)
2022	ACM China Doctoral Dissertation Award, Honorable Mention (Advisor: Yi-Chao Chen)
2022	Shanghai Computer Society Outstanding Doctoral Dissertation Award (Advisor: Yi-Chao Chen) (上海市計算機學會優秀博士學位論文獎、指導教師)
2022	China Communications Society Science and Technology Progress Award, Third Price. (中國通信學會科學技術獎、三等獎, 8/10)
2021	Shanghai Computer Society Science and Technology Invention Award, First Prize.

(上海市計算機學會科學技術發明獎、一等獎, 5/7)
 2021 Shanghai Computer Society Teaching Achievement Award, First Prize.
 (上海市計算機學會教學成果獎、一等獎, 6/10)
 2021 World Artificial Intelligence Conference 2021, Honorary Mention.
 2021 Techconnect Defense Innovation Awards.
 2019-2025 Shanghai Jiao Tong University MEM Program, Outstanding Teaching Award.
 2020 IEEE ComSoc Student Competition, Honorary Mention (Advisor: Yi-Chao Chen)
 2019 The 8th "Ecology-Value" Jinbo Award, Start-up Leader Award.
 2019 The Fifth Youth Innovation Competition, 2nd Prize.
 2019 SJTU Double-First Class Award.
 2018 CMU Summit on US-China Innovation and Entrepreneurship, Finalists.
 2017 NSF SBIR Phase II Award
 2017 7th Annual North America Chinese Startup Contest (UCAHP) 1st Prize.
 2014 ACM Travel Grant for MobiCom 2014, IMC 2014
 2011-2014 UT CS travel grant for MobiCom 2011, Infocom 2013, MobiCom 2014, IMC 2014, and etc.
 2011 Travel grant for S3 PhD forum organized with ACM MobiCom.
 2004 Scholarships of Taichung Special Municipality Congress Union
 2000-2003 Scholarships of Tai-Power Union